AC9M6N05 • Year 6 Mathematics

Adding and Subtracting Fractions

Use equivalent fractions, benchmarks and mixed-number interpretation

We are learning to…

Students generate common denominators, combine equal-sized parts, bridge whole numbers and interpret improper or mixed results in context.

Represent | Reason | Calculate | Interpret

Verify

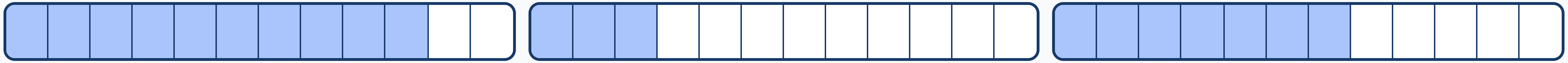
Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

solve problems involving addition and subtraction of fractions using knowledge of equivalent fractions

Solve 5/6 − 1/4 using equivalent fractions



5/6 = 10/12 | 1/4 = 3/12 | difference = 7/12

The denominator remains 12 because the result counts twelfths. Simplify only when numerator and denominator share a factor.

Choose fraction strategies by relationship

3/4 + 5/8

rename 3/4 as 6/8

7/10 + 3/10

complete one whole

1 1/3 − 5/6

rename whole and thirds as sixths

5/12 + 1/3

rename 1/3 as 4/12

Use diagrams or number lines when crossing a whole or interpreting a word problem; use factor knowledge to select the least useful common denominator.

Build and annotate the model

Represent adding and subtracting fractions and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Numerators and denominators both added:** The common denominator names the unchanged part size.
- **Fractions combined before renaming:** Different denominators name different-sized parts.
- **Whole part lost during subtraction:** Regroup one whole as denominator-sized parts.
- **Final fraction simplified incorrectly:** Divide numerator and denominator by the same factor.

Quick check

- Solve 5/6 − 1/4.
- Add 3/4 + 5/8.
- Subtract 1 1/3 − 5/6.
- Select a common denominator.
- Interpret an improper result.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.